Robot Procedure 2016_014

Apr 14th

2016

Insulation & Grounding Guidelines for Welding Robot

[Applicable for all Arc Weld Robots with any Weld Power Source]

Interfacing Tips are as below:

- ✓ Teach pendant and teach pendant cable
  Keep teach pendant and teach pendant cable over 0.5m away from the object where welding electric current flow, such as welding cable, jigs, iron plate used to fix the jig. High frequency noise with welding electric current may cause trouble, if it is inducted directly to teach pendant or teach pendant cable.
- ✓ Ground cable
  Provide ground cables of robot (ground cable between mechanical unit and controller is connected in standard specification), welding machine, wire feeder and others individually, which cables are over class D. Ground securely using ground cable that is designated by welding machine between welding machine and jig. Use attached four-core power cable (7m) to connect power and ground to robot controller.
- ✓ Interface of USB and PCMCIA
  It may become impossible to use the memory card, the sensor, PC etc, which use the USB or PCMCIA interface during the welding due to noise.
- ✓ Primary input wiring
  Provide switch box of primary input for robot and welding machine individually.
  In case of using some welding machines, provide switch box involving input protection equipment designated by each welding machine individually.
- ✓ Water cooling unit
  Don't set water cooling unit on the top panel of welding machine.
- ✓ Others
  1. While welding in progress, close all doors and covers of robot controller and welding machine securely.
  2. The way of connecting cables and the way of grounding affect the toughness for noise of welding machine. So, need to modify the way of connecting cables and grounding in case of generating gap of welding position or alarm.
  3. High frequency noise of welding may affect the below equipment's and may cause electromagnetic trouble in the below listed:
     a) Input cable, output cable, telephone cable, etc.
     b) Radio, television etc.
     c) Computer or other controller
     d) Detectors for industrial purpose or safety device
     e) Pacemaker or hearing aid

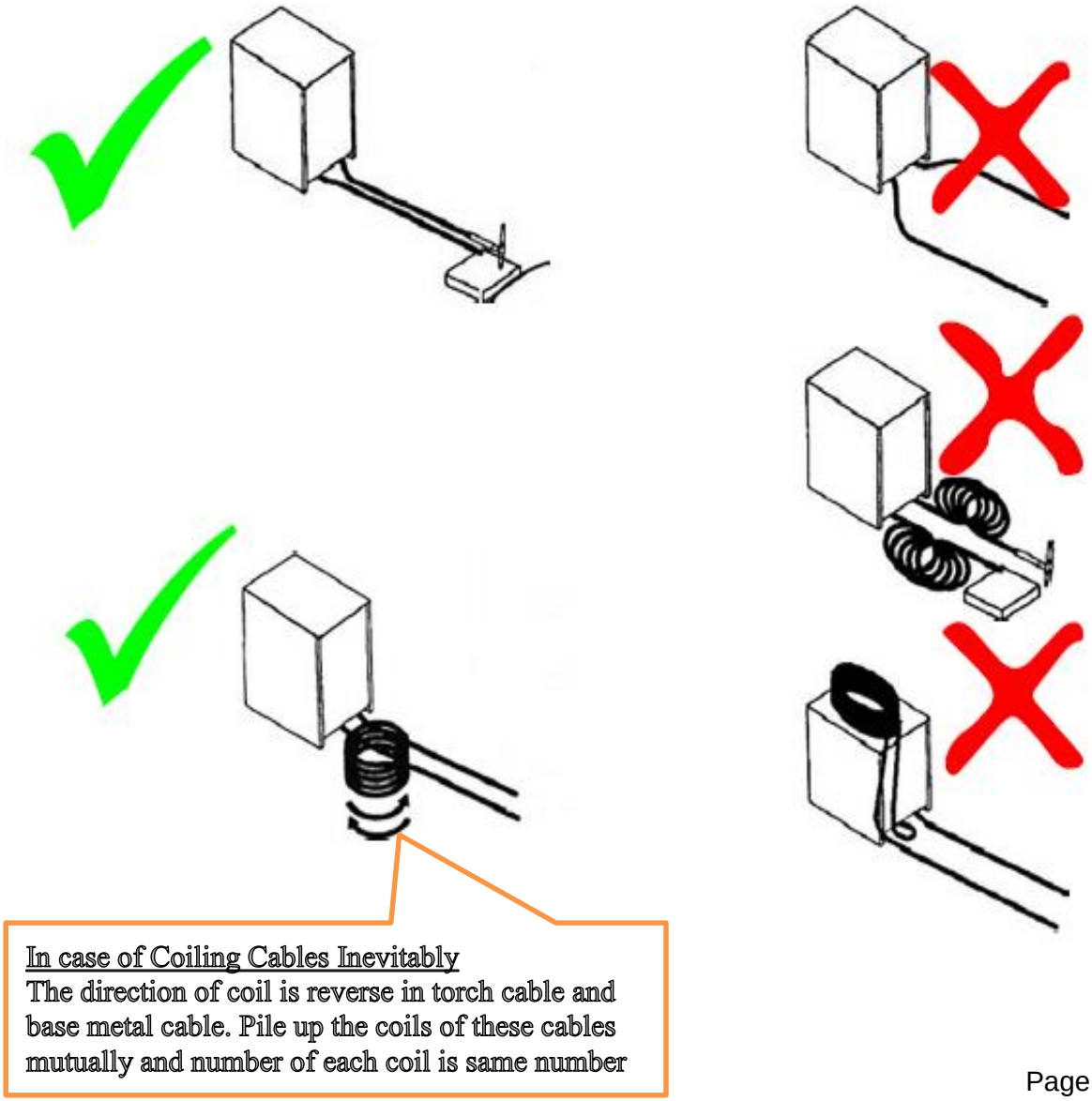
Make grounding in referring following matters.

- ✓ Use ground cable which conductor cross section size is over 8mm^2 as thick as possible and fix it securely.
- ✓ Construct a ground per a robot or a welding machine and not share with other equipment.
- ✓ Insulate robot mechanical unit from welding jig, welded work (plus side of welding machine) perfectly.
- ✓ Insulate welded work from jig body, auxiliary axis and robot common base (Insulation resistance should be over 1MΩ.) and insulate from robot controller. For inserting insulated material, use different bolts for insulated material mounting bolts and jig/outer axis/robot/robot controller mounting so that insulation is performed at the mounting part absolutely. Please contact your local FANUC representative for details.
- ✓ In case of using common base, construct ground that is beyond class D with cabtyre cable that is over 8mm^2 for common base and jig body. Don't share this ground with ground of robot or welding machine.
- ✓ In case of setting robot mechanical unit or controller on the common base, insulate them from common base surely.
- ✓ We recommend to connect the zipper tube of the torch cable to the earth of the high frequency unit of the welding machine
- ✓ Use base metal cable that is over 38mm^2 in order to prevent overheating of cable and to do proper welding. Strengthen the insulation effect of cables and prevent electric discharge among cables, robot body, iron plate fixing robot body and others
- ✓ Weld equipment connection cable, keep this cable away from arc generating zone, torch cable, base metal cable, base metal ground cable, input power cable, ground cable and others to prevent the troubles by inserting high frequency noise. Don't wire parallel to welding cable and robot connection cable as possible.
- ✓ Connect one side of ground cable to ground terminal of welding machine and do class C or class D grounding to another side of ground cable.

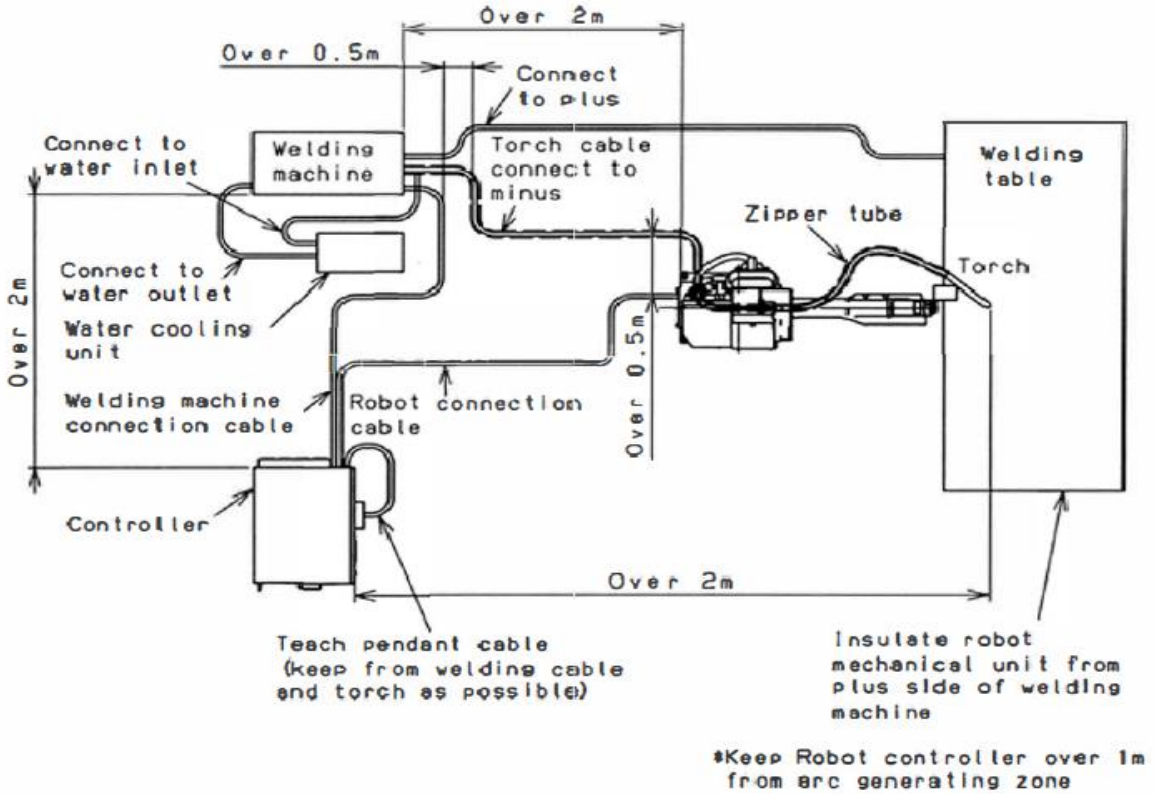
- ✓ Use ground cable which conductor size is more than one of input power cable (8mm^2 is needed as the lowest one)
- ✓ Grounding should be done by a person who has the license of electric engineering technician.
- ✓ Don't connect ground cable to water pipe or steel frame of building because it is not enough ground.
- ✓ The grounding information for Class C & D as illustrated as below:

Grounding	Description	Application
Class C	Grounding resistance should be 100 Ω or less. (For low voltage electric circuit, if the purpose is constructing the equipment that cut off its circuit automatically within 0.5 seconds when ground fault generate in its circuit, grounding resistance should be 500 Ω or less.)	Below 300V and low voltage
Class D	Grounding resistance should be 10Ω or less. (For low voltage electric circuit, if the purpose is constructing the equipment that cut off its circuit automatically within 0.5 seconds when ground fault generate in its circuit, grounding resistance should be 500 Ω or less.)	Over 300V and low voltage

- ✓ Please share this information with the Line Builder during Robot installation or troubleshooting for welding $+_{ve}$ leakage.
- ✓ Please refer the below sketch for detailed information on handling Weld Power Cable:

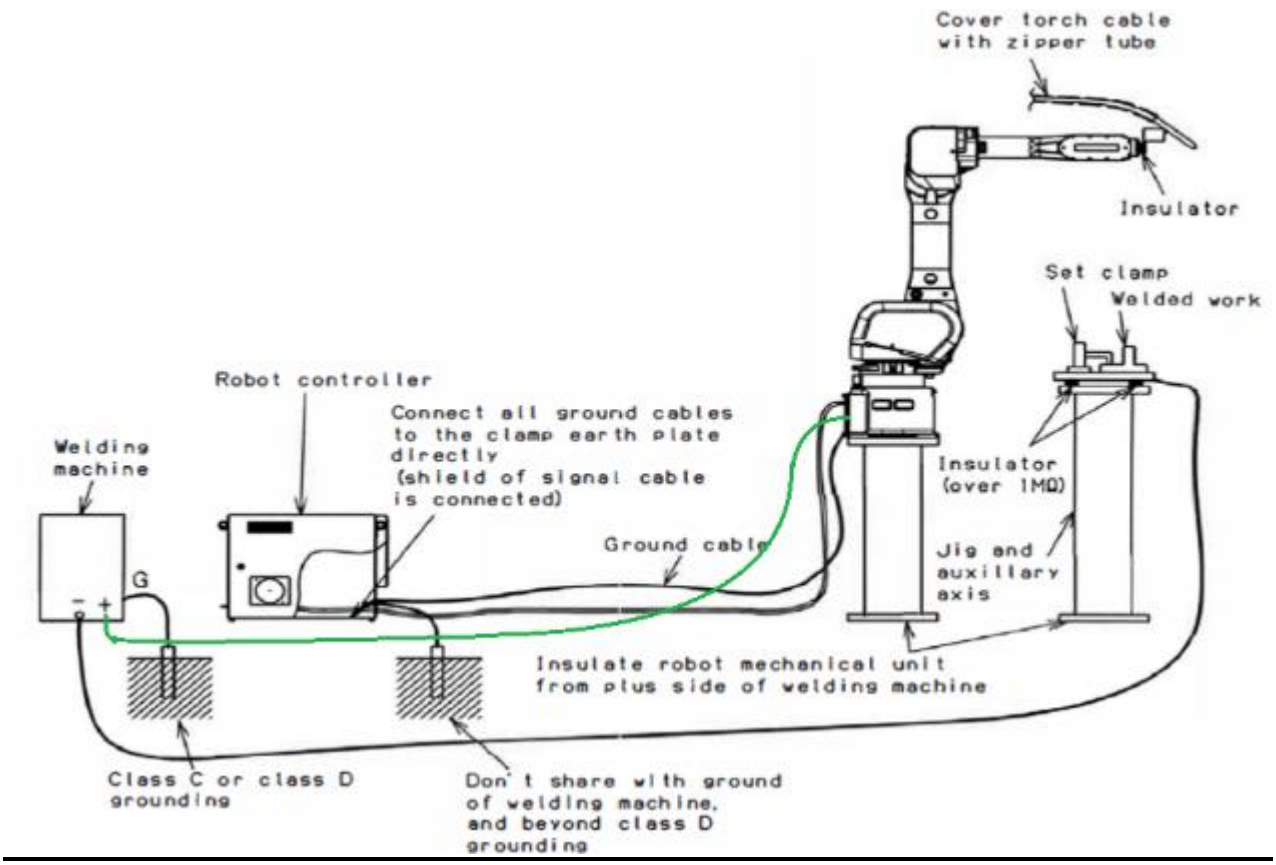


Cable Routing Sketch for Welding and Interfacing Cable Routing

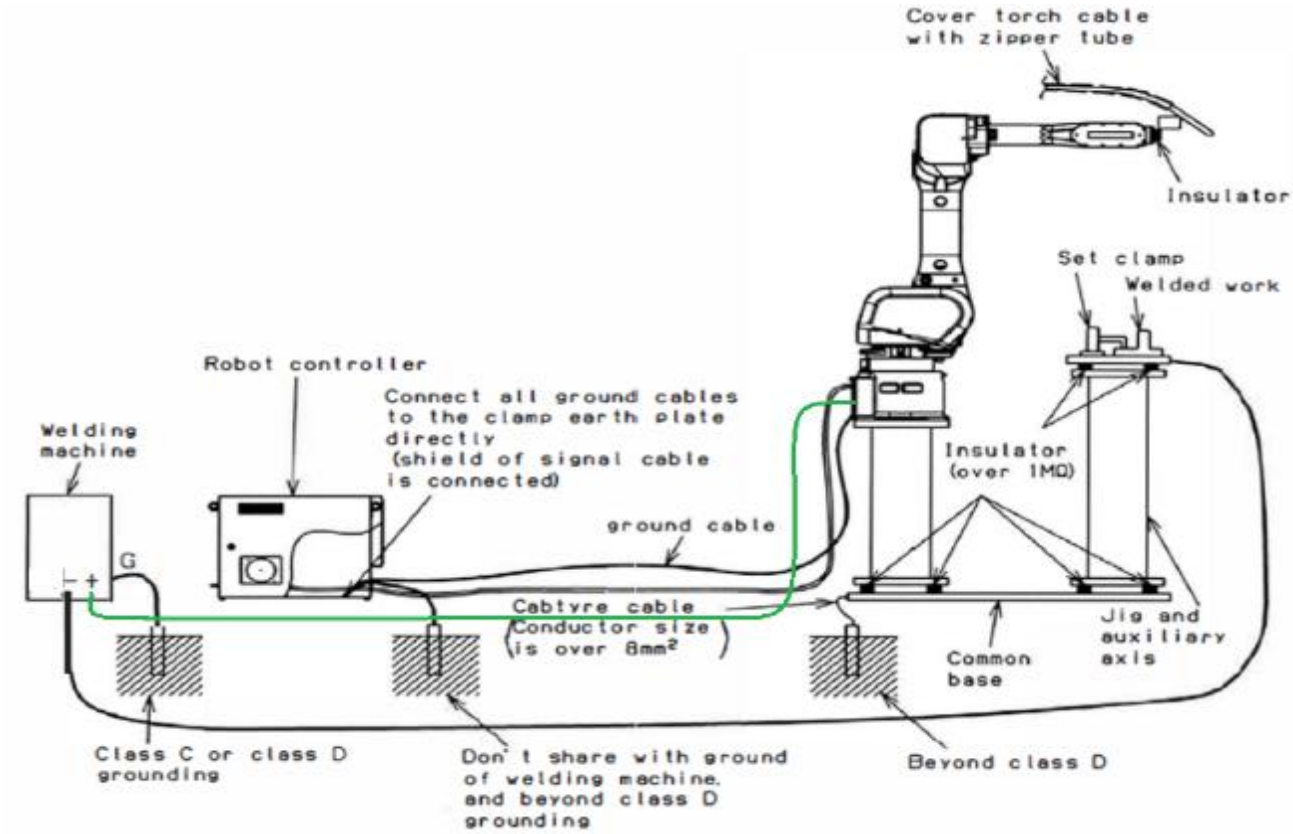


The above sketch notice of "Over 2m between Welding machine and Controller" does not needed for Lincoln Power Wave i400 that is integral type of Welding machine and Controller.

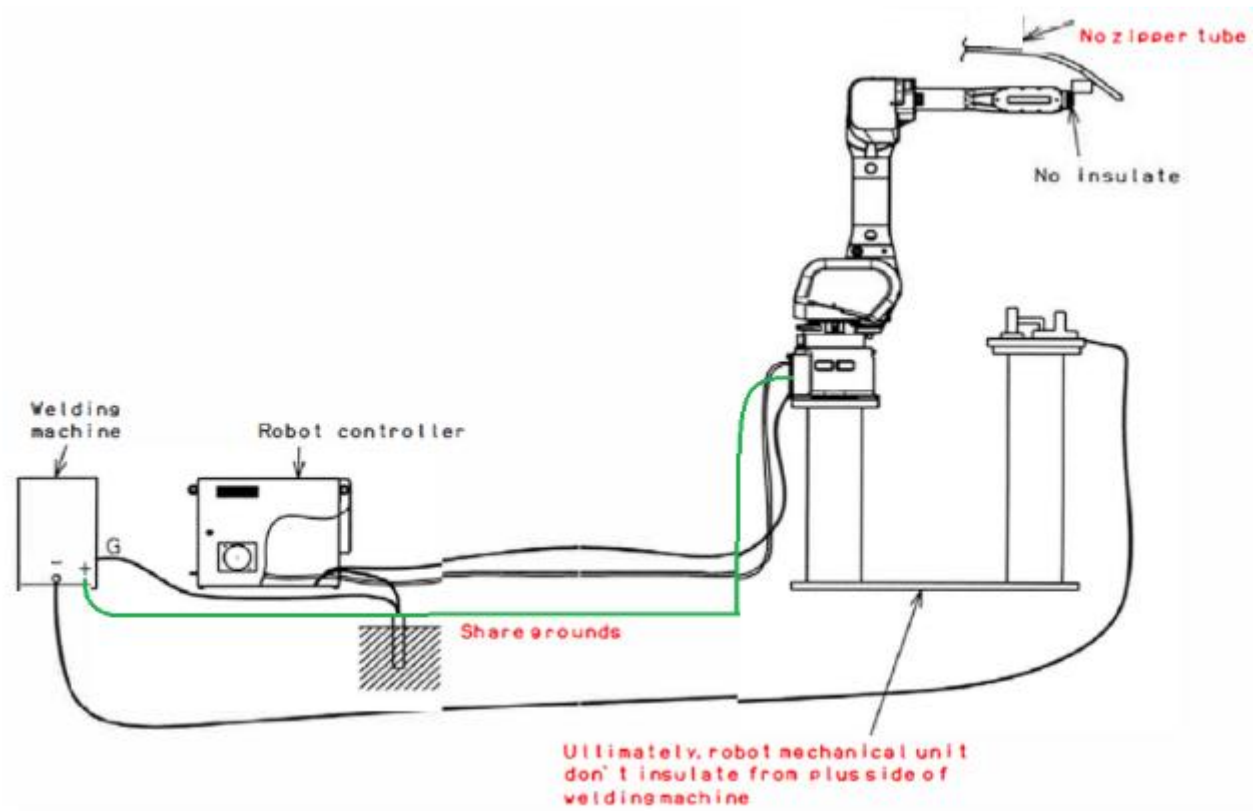
Example for Good Grounding and Insulation (No Common Base)

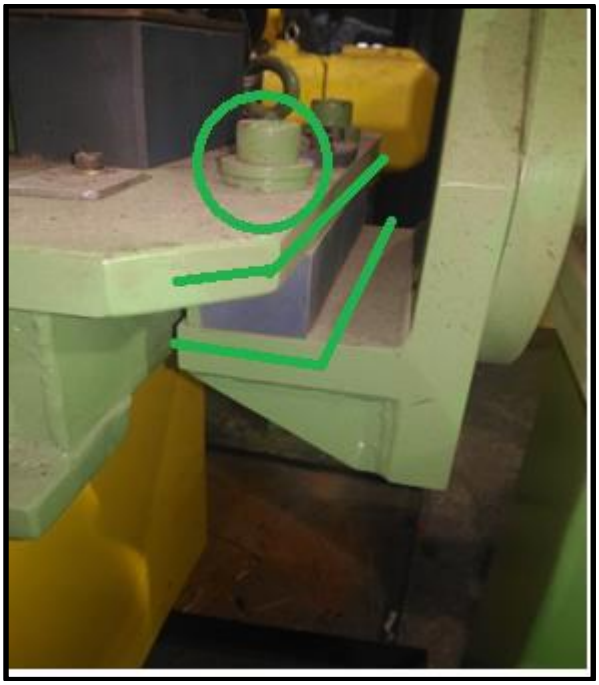


Example for Good Grounding and Insulation (Common Base)



Example for Bad Grounding and Insulation (No Common Base)



Bad example for Robot Base not insulated (Need to insulated areas are highlight with **Red** Colour marking:

Good Example for Extended Servo Axis Motor & Fixture Support mounting Insulated (Insulated Area are highlighted with **Green** colour line marking: